

Functional Requirements Document

Project: DwellEdge POS for Ice Cream Outlet (Multi-Tenant Architecture)

Version:	1.2
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Status:	Draft with Multi-Tenant & Multi-Outlet Architecture

1. System Overview

A dynamic, multi-tenant capable desktop POS application designed for multi-client and multi-outlet ice cream billing ecosystems. While operating with 100% offline edge capabilities for rapid checkout performance, the system dynamically configures its entire behavior, branding, item masters, and Oraxey ERP endpoints based on cloud-validated cryptographic licenses. Designed explicitly for high-speed counter use with touch screens, supporting both scoop-based and precise weight-based selling.

2. Functional Modules

Module	Description	Priority
2.1 Login & Security	Staff PIN login, Manager PIN for settings/refund/day close. Central multi-tenant license validation on startup.	Must-have
2.2 Billing Screen	Dynamic touch grid adapting to localized flavor profiles, topping matrices, barcode scan, weight scale metrics, and customer cart processing.	Must-have
2.3 Item Search & Scan	Multi-index matching engine: Barcode scan → auto-add to cart Search by Name Search by Short Wrapper Item ID.	Must-have

2.4 Weight-based Selling	Real-time scale integration via RS232/USB. Auto price calculation = Localized Rate/kg × Measured Weight. Manual manager overrides.	Must-have
2.6 Payment Handling	Cash, Card, UPI. Split payment support. Variable local currency/change auto-calculation.	Must-have
2.7 Thermal Printing	58mm/80mm layout engine supporting dynamic receipt headers mapping directly to the authenticated client's branch metadata.	Must-have
2.8 Refund/Cancel	Localized bill lookup or barcode search. Transaction reversal requesting designated branch manager PIN verification.	Must-have
2.9 Day Close Z-Report	Shift-wise revenue metrics, tender breakdowns, data sync validation status packets compiled locally per register.	Must-have
2.10 Offline Sync Engine	Isolated local SQLite database ensuring 100% sandbox protection. Pushes transaction queues securely to client-specific ERP endpoints.	Must-have
2.11 Licensing & Bootstrapping	On first-run, contacts global provisioning server via license key. Downloads target server URLs, tenant configs, and	Must-have



3. Key Business Rules - Updated

3.1 Item Identification & Mapping

- **Actual Item ID:** Global ERP unique primary key (e.g., 1234333322). Utilized explicitly for backend ledger sync across databases.
- **Wrapper Item ID:** Tenant-scoped short code (2-3 digits, e.g., 241) optimized for high-speed physical counter selection.
- **Barcode Tracking:** EAN-13 structure for packaged units. Instantly queries the engine cache to populate active ticket rows.
- **Multi-Index Search Priority:** Barcode parsing > Wrapper ID match > Lexical Item Name query > Dynamic UI grid mapping.

3.2 Dynamic Selling Operational Modes

Selling Mode	Triggering Action	Pricing Matrix Logic	Receipt String Output (Sample)
Scoop-based	Touch interface flavor cell selection	Quantity × Tenant Flat Price Unit	2x Vanilla Scoop - ₹ 120.00
Weight-based	Target SKU activation + Automated Scale Feed	Measured Weight (kg) × Localized Rate/kg	0.250kg Vanilla @ ₹400.00/kg - ₹ 100.00
Packaged Goods	Hardware scanner barcode read input	Fixed Global Master Unit Value	1x Cornetto - ₹ 50.00

3.3 Wrapper ID Validation Framework

- **Synchronization Integrity:** The specific client ERP engine must broadcast both the downstream ActualID and its mapped local WrapperID.
- **Cashier Lookups:** Typing short codes into the main checkout prompt triggers instant resolution to full context descriptors.
- **Layout Requirements:** Receipt strings and report lines prepend the short code for rapid physical cross-referencing.
- **Exception Handling:** If code resolution fails, explicit user alerts are thrown: "Item [ID] not found in local master. Force cache synchronization?"

3.4 Multi-Tenant Architecture & Multi-Outlet Provisioning Rules

- **Zero-Hardcoding Framework:** The core app distribution binary is 100% agnostic of specific database structures, server endpoints, or branding strings.
- **Dynamic Endpoint Ingestion:** Upon initial handshake with the centralized

provisioning system, the software configures localized secure routing rules mapping to that precise tenant's ERP server cluster.

- **Data Isolation Sandboxing:** Local SQLite storage containers isolate all database files by tenant hash and outlet codes, ensuring complete zero-leakage cross-contamination protocols across business segments.
- **Branch Configuration Autonomy:** Local peripheral configurations (scale interfaces, printer drivers, local currency symbols) exist independently per installation node, allowing decentralized terminal setups under a single tenant dashboard.

4. UI/UX Requirements - Updated

- **Dynamic Terminal Prompt:** Search fields feature context-aware watermarks reflecting lookups by name, barcode, or configured short codes.
- **Quantized Measurement Overlay:** Weighable SKU actions launch a dedicated oversized virtual numpad complete with direct scale query execution triggers.
- **Line Item Clear Clarity:** Display grids explicitly detail short identifiers alongside computed metric weights for seamless checkout tracking.
- **Tenant Initialization Dashboard:** First-run screens intercept administrative focus, requesting secure key injection to resolve business branding and inventory parameters dynamically.

5. Integration Requirements - Oraxey ERP - Updated

Target API Component	Data Vector Direction	Execution Interval	Injected Protocol Metadata Parameters
Item Master Pipeline	Oraxey Tenant Cloud → Local App Cache	Authentication handshakes + explicit pull routines	ActualID, WrapperID, RatePerKG, BaseUOM, WeighableFlag, Tenant Hash
Secure Bill Sync Pipeline	Local Engine Queue → Target ERP Target	Real-time stream pipeline (Fallback to local batch queue)	BranchCode, TerminalID, WrapperID, NetWeight, TargetRate, ComputedGST
Global Licensing Handshake	Local Runtime Environment →	Application launch lifecycle	Cryptographic LicenseKey →

Licensing Auth Server	initialization	Returns Target ERP URL, Active Tenant Code, Feature Flags, Expiration Epoch
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6. Bootstrapping, Provisioning & Licensing Flow

Step 1: Deployment Installation — Administrators execute silent setups incorporating global license strings: IceCreamPOS_Setup.exe /silent /license=CLIENT-OUTLET-HASH

Step 2: Tenant Routing Discovery — The runtime contacts the global auth hub, decrypts the token parameter, and updates internal routes to point exclusively to the respective client's enterprise servers.

Step 3: Dynamic Context Seeding — The app instantiates the local SQLite instance, parsing custom corporate headers, tax schemas, and localized branch item structures.

Step 4: Endpoint Operational Handshake — Verifies system linkages and binds physical peripherals. UI loads customized operational contexts instantly.

Licensing Enforcement Policies: Upon validation failure or subscription block tracking, checkout mechanisms lock, restricting access exclusively to auditing and reports viewer contexts.

7. Hardware Requirements - Updated

Hardware Component	Minimum Performance Operational Blueprint Specifications	Target System Deployment Purpose
All-in-One Touch Terminal	15.6" High-Luminance Panel, Intel i3 Equivalent, 8GB DDR4 RAM, 256GB NVMe SSD	Local core computational engine execution host.
Heavy-Duty Thermal Printer	80mm / 58mm Formats, ESC/POS Emulation Native Compliance, High-Speed USB Interface	Customer physical receipt production engine.
Omnidirectional Barcode Scanner	USB HID Interface Profile, Hardware Support for Standard 1D and Dense 2D Codes	Instant packaged SKU resolution and receipt scans.

Integrated Weighing Scale	RS232 / USB Serial Pipeline Interface, 6kg Threshold Capacity, Verified 1g Precision Step	Real-time weight ingestion for loose items.
Electronic Cash Drawer	Heavy-Gauge Steel Enclosure, RJ11 Pulse Trigger to Thermal Printer Solenoid Kick	Secure localized currency containment handling.

8. Out of Scope - Phase 1

- Multi-terminal sync within outlet (Cross-register database broadcasting via mesh formats)
- Comprehensive Customer CRM engine modules
- Real-time Kitchen Display System (KDS) coordination clusters
- Centralized Corporate Mobile Dashboard applications for remote tenant owners
- Multi-variant UOM processing metrics (Strict baseline enforcement: 1 SKU maps uniquely to exactly 1 UOM value)

Immediate Strategic Next Steps:

1. Finalize architectural structural payloads for Oraxey API multi-tenant routing tables.
2. Audit vendor hardware protocols for selected weighing scale reference units to implement native device drivers.